Distributions of Random Variables

OpenIntro Statistics, Chapter 4

Recap: discrete vs. continuous random variables

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- A **continuous** random variable takes any value in an interval.

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Today we go deeper on the discrete side, using one running example throughout, then bridge back to the normal distribution we already know.

The Milgram experiment

Stanley Milgram (Yale, 1963) studied obedience to authority: an experimenter orders a "teacher" to shock a "learner" every time the learner answers incorrectly. (The shocks weren't real — the learner was an actor.) . . .

- Milgram found about **65%** of people would obey and administer the shock.
- Equivalently, **35%** refused.

Bernoulli random variable

Think of each participant as one trial with two outcomes. Call it a **success** if the person *refuses* to shock — so $p = 0.35$.

The geometric distribution

Dr. Smith wants to repeat this experiment, but only wants to sample people until she finds the ***first*** *one who refuses. What's the probability she stops after the 1st person? The 3rd? The 10th?*

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Geometric distribution

Describes the waiting time until the first success in independent, identically distributed (iid) Bernoulli trials:

$$P(\text{success on the } n\text{th trial}) = (1-p)^{n-1}p$$

Practice

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> *Can we find the probability of rolling a 6 for the first time on the 6th roll of a die using the geometric distribution?*

Yes — rolling a 6 is a clean success/failure outcome, one or the other happens every roll:

$$P(\text{6 on 6th roll}) = \left(\frac{5}{6}\right)^5 \left(\frac{1}{6}\right) \approx 0.067$$

Expected value of a geometric distribution

How many people is Dr. Smith expected to test before finding the first one who refuses?

$$\mu = \frac{1}{p} = \frac{1}{0.35} = 2.86$$

She's expected to test **2.86 people —** which raises a fair question: how can she test a non-whole number of people?

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What about the variance?

$$\sigma = \sqrt{\frac{1-p}{p^2}} = \sqrt{\frac{1-0.35}{0.35^2}} \approx 2.3$$

She's expected to test 2.86 people, give or take 2.3.

The binomial distribution: setup

> *Suppose we randomly select **4** individuals for the experiment. What's the probability that **exactly 1** refuses to shock?*

Call them Allen, Brittany, Caroline, Damian. Four different orderings satisfy "exactly 1 refuses":

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Scenario	A	B	C	D	Probability
1	refuse	shock	shock	shock	$0.35 \times 0.65^3 = 0.0961$
2	shock	refuse	shock	shock	0.0961
3	shock	shock	refuse	shock	0.0961
4	shock	shock	shock	refuse	0.0961

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3	shock	shock	refuse	shock	0.0961
4	shock	shock	shock	refuse	0.0961

Total: $4 \times 0.0961 = 0.3844$.

Counting scenarios without listing them all

Listing every ordering gets unmanageable fast — imagine $n = 9$, $k = 2$ refusals: `RRSSSSSSS`, `SRRSSSSSS`, `SSRRSSSSS`, … far too many to enumerate by hand.

The choose function

The number of ways to get exactly k successes in n trials:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

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- $\binom{4}{1} = 4$ — matches our hand count above.
- $\binom{9}{2} = 36$ — would've taken a while to list by hand! (In R: `choose(9, 2)`.)

The binomial formula

Binomial distribution

The probability of exactly k successes in n independent Bernoulli trials, each with success probability p:

$$P(k \text{ successes in } n \text{ trials}) = \binom{n}{k} p^k (1-p)^{n-k}$$

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(BINS) Binary Outcomes, Ind., Number Fixed, Same Sucess

Practice: Gallup obesity survey

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$$P(X = 8) = \binom{10}{8}(0.262)^8(0.738)^2 = 45 \times (0.262)^8 \times (0.738)^2 \approx 0.0005$$

Pretty low — makes sense, since 8-out-of-10 is far from the ∼2.6-out-of-10 we'd expect.

Expected value and variability

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$$\sigma = \sqrt{np(1-p)} = \sqrt{100 \times 0.262 \times 0.738} \approx 4.4$$

. . .

We'd expect 26.2 out of 100 to be obese, give or take 4.4 — even though 26.2 isn't a number of people that can literally occur in any single sample.

Unusual observations

Recall: observations more than 2 SD from the mean are considered unusual.

$$26.2 \pm (2 \times 4.4) = (17.4, 35)$$

Practice: A Gallup poll found 13% of Americans think home-schooling provides an excellent education. Would a sample of 1,000 people with only 100 sharing this view be unusual?

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$$\mu = 1000(0.13) = 130, \qquad \sigma = \sqrt{1000(0.13)(0.87)} \approx 10.6$$

How many deviations away is that?

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How many deviations away is that?

$$Z = \frac{100 - 130}{10.6} \approx -2.83$$

Yes — more than 2 SD below the mean, so unusual.

The standard normal distribution

Notice we've been converting to Z-scores every time we asked "is this unusual?" That conversion always lands on the same specific distribution:

Standard normal distribution

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- Any normal distribution can be converted to the standard normal by standardizing:

$$Z = \frac{x - \mu}{\sigma}$$

- Z tells you how many standard deviations x is above (positive) or below (negative) the mean — this is exactly what we computed for the home-schooling example ($Z \approx -2.83$).

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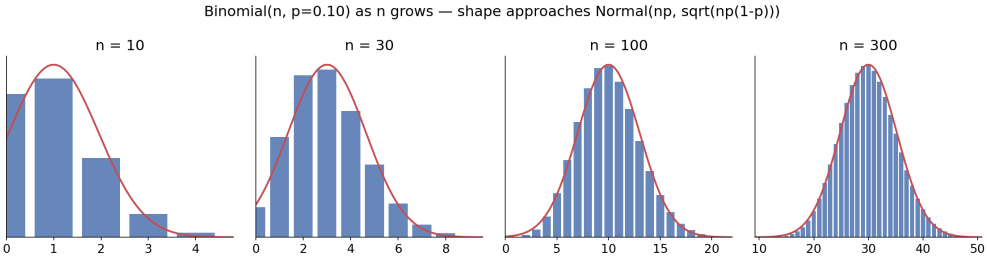
- Z tells you how many standard deviations x is above (positive) or below (negative) the mean — this is exactly what we computed for the home-schooling example ($Z \approx -2.83$).
- Why bother? Once something is standardized, we can look up (or compute with `pnorm()`) probabilities for **any** normal distribution using a **single** reference distribution, instead of a different table for every possible μ and σ.

What happens to the shape as n grows?



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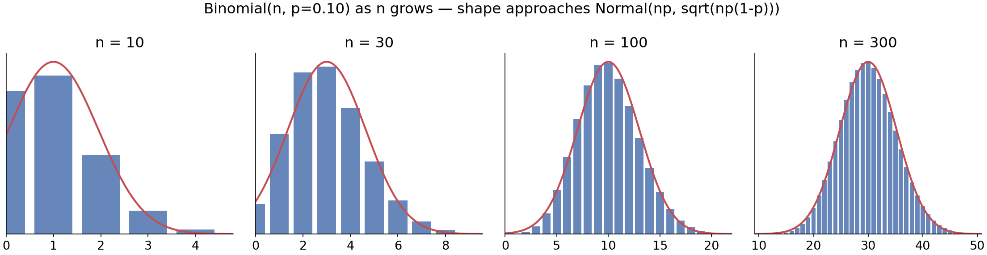

At $n = 10$, $p = 0.10$, the distribution is noticeably right-skewed. By $n = 300$, it's nearly symmetric and bell-shaped.

How large is "large enough"?

Condition for the normal approximation

The sample size is large enough to approximate a binomial with a normal distribution when **both**:

$$np \geq 10 \qquad \text{and} \qquad n(1-p) \geq 10$$

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Check: for $n = 10, p = 0.13$: $np = 1.3$ — **fails**, too skewed to approximate.

For $n = 25, p = 0.45$: $25(0.45) = 11.25$ and $25(0.55) = 13.75$ — **passes**.

Facebook power users

A Pew study found Facebook users "get more than they give" (more friend requests received than sent, more likes received than given, etc.) — explained by **power users** who contribute disproportionately more content.

$\sim$25% of users are power users; the average user has 245 friends.

*What's the probability that the average user (245 friends) has **70 or more** power-user friends?*

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~25% of users are power users; the average user has 245 friends.

*What's the probability that the average user (245 friends) has **70 or more** power-user friends?*

Directly: $P(X \geq 70) = P(X = 70) + P(X = 71) + \cdots + P(X = 245)$ — a lot of binomial terms to add up.

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Check conditions:

$np = 61.25 \geq 10$, $n(1-p) = 183.75 \geq 10$.

What's the mean and variance?

Solving it with the normal approximation

What parameters do we know?

$n = 245$, $p = 0.25$.

Check conditions:

$np = 61.25 \geq 10$, $n(1-p) = 183.75 \geq 10$.

What's the mean and variance?

$$\mu = 245(0.25) = 61.25, \qquad \sigma = \sqrt{245(0.25)(0.75)} \approx 6.78$$

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$$\text{Bin}(245, 0.25) \approx N(61.25,\ 6.78)$$

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$$Z = \frac{70 - 61.25}{6.78} \approx 1.29$$

. . .

$$P(Z > 1.29) \approx 0.0985$$

In R: `1 - pnorm(1.29)`. About a 9.85% chance — much easier than summing 176 binomial terms by hand.

A word of caution

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- Fix: extend the cutoff by **0.5 in each direction** (a **continuity correction**) when estimating a range of counts. Option in R $\sim$
- This matters most for narrow ranges; for wide tail areas like the Facebook example, the correction barely changes the answer.

Why this matters going forward

A **sample proportion** $\hat{p}$ is really a rescaled binomial count — successes divided by n. Because binomial counts are approximately normal for large n, sample proportions are too.

Point estimates and the Central Limit Theorem

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Polling organizations like Pew Research survey samples to estimate opinions of an entire population.

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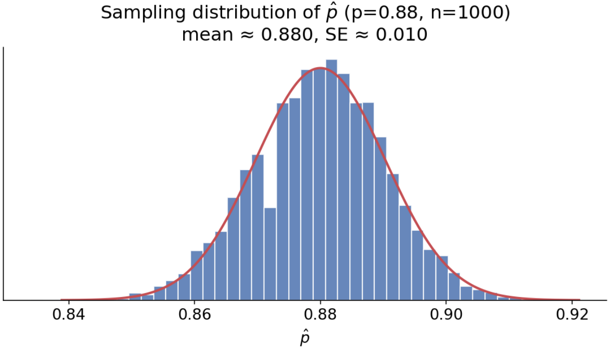
- **Sampling error** (a.k.a. sampling variability): how much $\hat{p}$ naturally bounces around from sample to sample.
- **Bias**: a *systematic* tendency to over- or under-estimate — e.g., from a leading survey question. Good data collection minimizes this; it's not something a bigger sample fixes.

Understanding sampling variability

Suppose the true proportion of American adults who support expanding solar energy is $p = 0.88$. If we poll 1000 people, how close would we expect $\hat{p}$ to land to 0.88?

We can simulate this on a computer — draw a random sample of 1000 "adults" from a population where 88% support solar, and compute $\hat{p}$. Repeat 10,000 times:



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- **Center:** essentially $p = 0.88$ — the sampling process is unbiased.
- **Spread:** the standard deviation of this distribution has a special name — the **standard error**, $SE_{\hat{p}}$.
- **Shape:** symmetric, bell-shaped — it *resembles a normal distribution*.

The Central Limit Theorem

Central Limit Theorem (for a proportion)

When observations are independent and the sample size is sufficiently large, $\hat{p}$ follows an approximately normal distribution:

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Applying the CLT to real poll data

Pew's actual solar-energy poll: $n = 1000$, observed $\hat{p} = 0.887$ (we don't know the true p — that's the whole point of polling!).

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- **Independence:** simple random sample
- **Success-failure condition:** since we don't know p, substitute $\hat{p}$ (called the **plug-in principle**):

$$n\hat{p} = 1000(0.887) = 887 \geq 10, \qquad n(1-\hat{p}) = 113 \geq 10 \quad \checkmark$$

. . .

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The CLT applies — we can model $\hat{p}$ as approximately $N(0.887, 0.010)$.

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Using $N(\mu_{\hat{p}} = 0.88, SE_{\hat{p}} = 0.010)$*: what fraction of samples would give* $\hat{p}$ *between 0.86 and 0.90?*

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$$Z_{0.86} = \frac{0.86 - 0.88}{0.010} = -2, \qquad Z_{0.90} = \frac{0.90 - 0.88}{0.010} = 2$$

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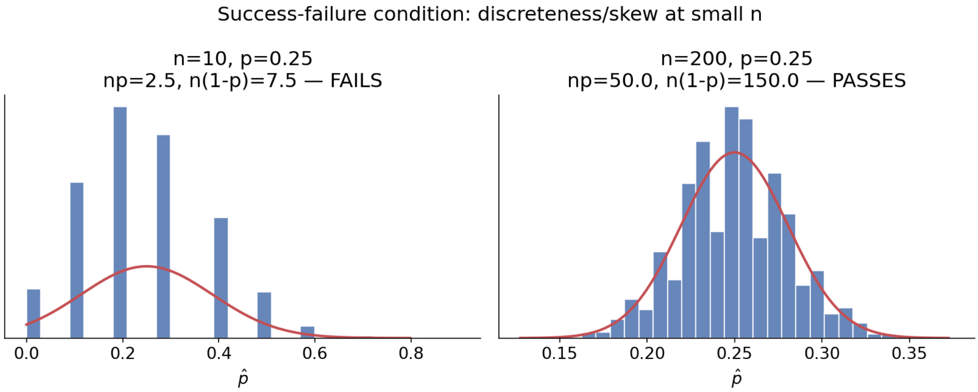
Tail areas are each 0.0228, so the area between is $1 - 2(0.0228) = 0.9544$.

About 95.4% of samples of size 1000 would give a $\hat{p}$ within 0.02 of the true proportion — this is exactly the logic behind a margin of error.

When the success-failure condition fails

$p = 0.25$, but now with $n = 10$ (fails: $np = 2.5$) vs. $n = 200$ (passes: $np = 50$):



At small n, $\hat{p}$ can only take a few discrete values (multiples of $1/n$) and the distribution is visibly skewed — the normal curve is a poor fit. The larger n gets, the smoother and more symmetric it becomes.

The CLT for a sample mean

Everything we just did with $\hat{p}$ has a direct parallel for a **sample mean** $\bar{x}$ estimating a population mean μ — say, average commute time instead of a yes/no proportion.

Central Limit Theorem (for a mean)

When observations are independent and the sample size is sufficiently large, $\bar{x}$ follows an approximately normal distribution:

$$\mu_{\bar{x}} = \mu, \qquad SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

where σ is the population standard deviation.

Cont.

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- "Sufficiently large" here doesn't use the success-failure condition (there's no "success/failure" for a numerical variable) — instead we typically want $n \geq 30$, *or* a population that's already approximately normal to begin with.
- Remarkably, this holds **almost regardless of the shape of the original population distribution** — even if individual observations are skewed or weird, $\bar{x}$ still tends toward normal as n grows. This is really *why* the theorem is called "central" — it's a hub result that keeps reappearing throughout statistics.